



PUCHA ARUN KUMAR

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	Master of Technology	Indian Institute of Technology, Kharagpur	7.48 / 10
2025	Bachelor of Technology	Gayatri Vidya Parishad College for Degree and PG Courses	8.37 / 10
2021	Board of Intermediate Education, AP	Sri Chaitanya Junior College	87.4 %
2019	Central Board of Secondary Education	Kendriya Vidyalaya Malkapuram	78.8 %

COMPETITION/CONFERENCE

Paper Title: AI-Driven 3D Blueprint Generator for Personalized Product Design
Conference Publication (ICRRCE-2025) - Accepted for Scopus-indexed Bentham Books

Architected an end-to-end generative AI pipeline converting text prompts into manufacturable 3D blueprints. Implemented multi-view image generation using DALL-E 3 and Stable Diffusion XL for design consistency. Applied Neural Radiance Fields and 3D Gaussian Splatting with SLAT for accurate 3D reconstruction. Developed automated mesh reconstruction workflows using FreeCAD and Blender APIs for CAD outputs.

INTERNSHIPS

Machine Learning Engineer Intern | Aegion Dynamic Solutions, Visakhapatnam | [Link](#) **May 2024 - August 2024**

Developed NLP-based inference and topic clustering models for legal contracts to improve retrieval accuracy. Benchmarked transformer models, including BERT, RoBERTa, and DeBERTa, across multiple NLI datasets. Evaluated cross-domain generalization using MultiNLI, Legal-NLI, SNLI, and task-specific benchmark datasets. Enhanced document analysis pipelines and contributed to frontend architecture for the Nimbus SaaS platform.

SKILLS AND EXPERTISE

Programming & ML Frameworks: Python | PyTorch | TensorFlow | Scikit-learn | Keras | Hugging Face
Core ML Skills: Machine Learning | Deep Learning | NLP | Neural Networks | Model Optimization
Data & Analysis: NumPy | Pandas | Matplotlib | Seaborn | EDA | Data Preprocessing & Visualization
MLOps & Systems: FastAPI | MLflow | Docker | SQL | Vector Databases | Git | Azure

COURSEWORK INFORMATION

Artificial Intelligence | Machine Learning | Deep Learning | Natural Language Processing | Computer Vision | Optimization Techniques | Reinforcement Learning | Data Structures and Algorithms | Data Engineering

PROJECTS

Retinal Disease Detection

Developed a fusion-based deep learning model combining CLIP and DenseNet-121 for retinal analysis. Built an automated diabetic retinopathy grading system using the APTOS 2019 real-world dataset. Applied robust data preprocessing and augmentation techniques to handle real-world image variations. Achieved 85.45% accuracy and 0.92 Quadratic Weighted Kappa, reflecting strong clinical agreement. Integrated Grad-CAM based explainability to highlight lesion-specific regions and support clinical trust.

Skin Disease Detection

Developed a hybrid CNN using ResNet50 and DenseNet121 for multi-class skin disease classification. Trained and evaluated the model on a real-world dataset with 5,000+ images across 22 disease classes. Achieved 79.11% test accuracy, outperforming baseline CNN models through feature fusion techniques. Integrated OOD detection and Grad-CAM explainability to enhance model safety and interpretability.

Fake News Detection using BERT for binary text classification

Implemented a custom BERT-based transformer architecture in PyTorch for fake news classification. Preprocessed and balanced the Fakeddit dataset with 180k+ samples using unbiased train-test splits. Handled hierarchical post-comment structures to preserve contextual dependencies during training. Achieved 84% test accuracy with a 0.83 F1-score using AdamW optimization and contextual embeddings. Evaluated model generalization using transformer-based representations on unseen news samples.

Residual RNN-Based Sequence Modeling for Code Completion

Implemented RNN, LSTM, and Residual RNN architectures in PyTorch for token-level code completion. Benchmarked models using Test Top-5 Accuracy and Perplexity to evaluate sequence modeling performance. Achieved best results with Residual RNN, improving accuracy by 1.22% and reducing perplexity by 8.08%. Demonstrated improved gradient flow and generalization using residual connections in recurrent networks.

EXTRA CURRICULAR ACTIVITIES

Research-oriented work in assistive technology and the medical domain involving AI model development, paper analysis, benchmarking studies, and structured technical documentation.